United States Patent Application Publication	20250276746
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	Krug; Stephen Leo et al.

SEMI-TRACTOR ROLLOVER PREVENTION DEVICE

Abstract

A rollover prevention device couples a semi-trailer to a semi-tractor. The rollover prevention device comprises a top plate configured to be coupled to the semi-trailer and a bottom plate configured to be coupled to the semi-tractor. One or more frangible fasteners couple the top plate to the bottom plate. The rollover prevention device includes a controller coupled to one or more sensors, with the controller configured to determine a roll angle and a roll rate of the rollover prevention device. In response to determining the roll angle and/or the roll rate satisfy a rollover condition, the controller transmits a control signal to separable fasteners, which separate in response to the control signal allowing the top plate to detach from the bottom plate.

Inventors: **Krug; Stephen Leo (Inglewood, CA), Walker; Kyle (Alamo Heights, TX)**

Applicant: **Axicle, Inc.** (Inglewood, CA)

Family ID: **96881810**

Appl. No.: **19/069213**

Filed: **March 03, 2025**

Related U.S. Application Data

us-provisional-application US 63561220 20240304

Publication Classification

Int. Cl.: **B62D49/08** (20060101); **B62D53/08** (20060101)

U.S. Cl.:

CPC **B62D49/08** (20130101); B62D53/08 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/561,220, filed Mar. 4, 2024, which is incorporated by reference in its entirety.

BACKGROUND

[0002] A semi-trailer is attached to a semi-tractor through a fifth wheel connection. As the semi-tractor moves, the semi-trailer also moves. However, the center of gravity of a semi-tractor is significantly lower than the center of gravity of a semi-trailer. Therefore, a semi-tractor without an attached trailer is more resistant to rollovers than a semi-tractor with an attached trailer. Because of this, most rollovers of a semi-tractor are caused when a semi-trailer is attached to the semi-tractor. Often, a rollover of a semi-tractor with an attached semi-trailer is caused by the torque that the semi-trailer applies to the semi-tractor the fifth wheel connection during high-speed cornering, high winds, or while driving on a highly inclined surface.

[0003] About half of all semi-tractor drivers that die in on-road accidents die when the semi-tractor rolls over. Existing solutions such as antilock brakes and electronic stability control have not significantly reduced the number of rollover accidents that occur each year.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** is a perspective view of a location on a semi-tractor where a fifth wheel is typically installed, in accordance with one or more embodiments.

[0005] FIG. **2** is a top perspective view of an example of a rollover prevention device with a top plate installed, in accordance with one or more embodiments.

[0006] FIG. **3** is a side cross-section view of an example of a rollover prevention device with a top plate installed, in accordance with one or more embodiments.

[0007] FIG. **4** is a top perspective view of a rollover prevention device without a top plate installed, in accordance with one or more embodiments.

[0008] FIG. **5** is a block diagram of an architecture of a controller of a rollover prevention device, in accordance with one or more embodiments.

[0009] FIG. **6** is a side cross-section view of an example rollover prevention device without a top plate installed, in accordance with one or more embodiments.

[0010] FIG. **7** illustrates an example of a top plate of the rollover prevention device pivoting off of the mounting cones and bottom plate of the rollover prevention device, in accordance with one or more embodiments.

[0011] FIG. **8** illustrates one embodiment of assembling pillow blocks of the rollover prevention device onto mounting cones of the rollover prevention device, in accordance with one or more embodiments.

[0012] FIG. **9** is a top perspective view of an alternative embodiment of a pillow block and a bottom plate of a rollover prevention device.

[0013] FIG. **10** is a side cross-sectional view of the alternative embodiment of the pillow block and the bottom plate of a rollover prevention device shown by FIG. **9**.

[0014] FIG. **11** is an overhead view of one embodiment of a repositionable fifth wheel connector.

[0015] FIG. **12** is an overhead view of an alternative embodiment of a repositionable fifth wheel connector.

[0016] FIGS. **13**A and **13**B are perspective views of another embodiment of a repositionable fifth wheel connector.

[0017] FIG. **14** is a flowchart of one embodiment of a method for determining whether to transmit a control signal to frangible fasteners of a rollover prevention device.

DETAILED DESCRIPTION

[0018] The Figures (FIGS.) and the following description describe certain embodiments by way of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein. Reference will now be made to several embodiments, examples of which are illustrated in the accompanying figures. Wherever practicable, similar or like reference numbers may be used in the figures and may indicate similar or like functionality.

[0019] The disclosed embodiments include a rollover prevention device configured to couple a semi-trailer to a semi-tractor. The rollover prevention device includes a top plate and a bottom plate. The top plate is configured to be coupled to a semi-trailer, and the bottom plate is configured to be coupled to a portion of the semi-tractor. One or more frangible fasteners couple the top plate to the bottom plate. In various embodiments, one or more secondary securing mechanisms further couple the top plate to the bottom plate.

[0020] The rollover prevention device also includes one or more sensors communicatively coupled to a controller. In various embodiments, the sensors monitor one or more of: inertia, force, linear speed, rotational speed, vibration, or GPS signals. Based on data monitored by the sensors, the controller generates an estimate of the semi-tractor or the semi-trailer's attitude, heading, and/or global position. The attitude includes velocity, roll, pitch, yaw, angle estimates, rotational rates, rotational accelerations, and linear accelerations. The controller also includes, or is coupled to, an inertial measurement unit (IMU) comprising one or more gyroscopes or one or more multi-axis accelerometers. One or more of the sensors capture data capable of determining a roll angle of the rollover prevention device and a roll rate of the rollover prevention device.

[0021] In various embodiments, the controller and IMU, as well as one or more other systems, are included in a rollover prevention device and are positioned proximate to a fifth-wheel that connects the semi-trailer to the semi-tractor. However, in other embodiments the controller is proximate to the fifth-wheel connection, while one or more sensors communicatively coupled to the controller are in different positions of the semi-tractor or the semi-trailer. Based on data captured by one or more sensors, the controller determines whether a roll angle, a roll rate, and/or other conditions of the rollover prevention device satisfy a rollover condition. An example rollover condition comprises a roll angle of the rollover prevention device being within a threshold amount of a threshold roll angle. Another example rollover condition comprises a roll rate of the rollover prevention device equaling or exceeding a threshold roll rate. As another example, the rollover condition comprises a combination of a threshold roll angle and a threshold roll rate meeting predefined criteria. For example, the controller may perform a lookup of a determined roll angle and determined roll rate in a lookup table that maps these conditions to an output decision indicative of whether or not the rollover condition is met. In response to determining the roll angle and/or the roll rate satisfies a rollover condition, the controller transmits a control signal to the one or more separable (e.g., frangible) fasteners of the rollover prevention device to cause the fasteners to separate (e.g., by detonating frangible fasteners), thereby separating the semi-tractor from the semi-trailer. In other embodiments, the controller determines one or more rollover conditions based on alternative data obtained by the controller, such as a height of a center of gravity of a semi-trailer coupled to the rollover prevention device, a mass of a semi-trailer coupled to the rollover prevention device, or other data or combinations of data obtained by the controller. In other embodiments, one or more rollover conditions comprise one or more combinations of threshold values of a combination of data measured by one or more sensors coupled to the controller or data determined by the controller based on data captured by one or more sensors. Different embodiments may include different types of data in one or more rollover conditions.

[0022] In response to receiving the control signal, a separation fastener separates, allowing the top plate of the rollover prevention device to detach from the bottom plate of the rollover prevention device. For example, a separation fastener is a frangible bolt, and explosive charges in the frangible bolt detonate in response to the control signal. As the top plate is used to couple the semi-trailer to the semi-tractor, detaching the top plate allows the semi-trailer to detach from the semi-tractor. Because the center of gravity of the semi-trailer is higher than the center of gravity of the semi-tractor, detaching the semi-trailer from the semi-tractor causes the semi-tractor to self-correct even if the semi-trailer continues to rollover after detaching from the semi-tractor.
[0023] Other embodiments may include a controller for fifth wheel connector that does not necessarily include the separable fasteners. In this embodiment, the controller is coupled to one or more sensors, or includes one or more sensors, as further described above. Based on data obtained from one or more of the sensors, the controller determines one or more different quantities. For example, a controller determines a roll rate and/or a roll angle of the fifth wheel based on data received from one or more accelerometers or one or more gyroscopes. The sensed data may be provided to a driver of the semi-tractor, to a fleet administrator, or may be applied to other control instrumentation associated with the semi-tractor and/or semi-trailer.
[0024] In another embodiment, a fifth wheel connector (that may or may not include the separable fasteners for rollover prevention) includes an actuator or may be coupled to an actuator associated with a guiding system having a lower plate configured to be coupled to a surface of a semi-tractor. A controller included in the fifth wheel connector provides one or more movement signals to the actuator, which causes movement of the fifth wheel connector along the guiding system. For example, repositioning the fifth wheel connector along the guiding system via the actuator is used in combination with control of brakes for wheels to reposition the suspension of the semi-trailer relative to the semi-tractor By repositioning the suspension of the semi-trailer, mass is redistributed between axles of the semi-trailer suspension and axles of the semi-tractor suspension to comply with axle load specifications or other factors. This slider system may be used in embodiments in which the fifth wheel connector includes the separable fasteners for rollover prevention or in conjunction with a passive fifth wheel connector that does not necessarily include the separable fasteners.
[0025] FIG. **1** is a perspective view of an example location on a semi-tractor **100** where a fifth wheel **105** is typically installed. The fifth wheel **105** is a coupling device positioned at a rear of the semi-tractor. The fifth wheel **105** includes a horseshoe-shaped opening through which a kingpin protruding from a bottom front surface of a semi-trailer is inserted. When a semi-trailer is coupled to the semi-tractor **100** through the fifth wheel **105**, as the semi-tractor **100** turns, a surface of the semi-trailer having the kingpin at its center rotates against a top surface of the fixed fifth wheel **105**, which does not rotate.
[0026] FIG. **2** is a top perspective view of one embodiment of a rollover prevention device **200**. As shown in FIG. **2**, the rollover prevention device **200** comprises a top plate **205** that is coupled to a bottom plate **210**. The rollover prevention device **200** is positioned at location of the fifth wheel **105** of the semi-tractor **100** in various embodiments. The top plate **205** includes a horseshoe-shaped opening configured to receive a kingpin of a semi-trailer. The bottom plate **210** is configured to be coupled to a portion of a semi-tractor **100**. In various embodiments, the bottom plate **210** is bolted to a surface of the semi-tractor **100**, while in other embodiments the bottom plate **210** is coupled to the surface of the semi-tractor **100** using one or more alternative or additional coupling mechanisms.
[0027] FIG. **3** is a side cross-section view of an embodiment of the rollover prevention device **200** with the top plate **205** installed. As shown in FIG. **3**, the top plate **205** is coupled to one or more pillow blocks **305** and the bottom plate **210** includes one or more mounting cones **310**. The mounting cones **310** are rigidly attached to the bottom plate **210**. Example mechanisms for attaching the mounting cones **310** to the bottom plate **210** include welding the mounting cones **310**

to the bottom plate or bolting the mounting cones **310** to the bottom plate **210**.

[0028] A mounting cone **310** extends from the bottom plate **210** into a cavity within a pillow block **305**. Each mounting cone **310** is inserted into a cavity in a corresponding pillow block **305**. Inserting mounting cones **310** into cavities of corresponding pillow blocks **305** prevents the fifth wheel and semi-trailer from laterally or longitudinally sliding off the semi-tractor under non-rollover conditions. With the mounting cones **310** inserted into cavities of the pillow blocks **305**, the top plate **205** is lifted vertically from the bottom plate **210** or the top plate **205** is vertically pivoted about a portion of the bottom plate **210**, as further described above in conjunction with FIG. **6**. Additionally, having the mounting cones **310** insert into cavities on corresponding pillow blocks **305** reduces a likelihood of a semi-trailer coupled to the top plate **205** moving forward and into a rear of a semi-tractor to which the bottom plate **210** is coupled.

[0029] Additionally, a pillow block **305** is further coupled to the bottom plate **210** through one or more separation fasteners **315**. In various embodiments, a separation fastener **315** comprises a frangible bolt configured to separate into multiple pieces in response to receiving a control signal. Various mechanisms may be used for the separation fastener **315** to separate into multiple pieces in various embodiments. In various embodiments, a separation fastener **315** is inserted through the mounting cone **310** into a portion of a pillow block **305** having a cavity into which the mounting cone is inserted **310**. A separation fastener **315** is inserted into the mounting cone **310**, where the head of the separation fastener **315** is proximate to the bottom plate **210** in various embodiments. The pillow block **305** includes a threaded hole having internal threads that interlock with threads of the separation fastener **315** when the separation fastener **315** is inserted into the threaded hold of the pillow block **305**.

[0030] A frangible bolt may include one or more explosive charges capable of being detonated in response to a control signal. For example, the frangible bolt has a partially hollow shaft filled with an explosive material. In various embodiments, the frangible bolt receives a control signal to detonate one or more of the explosive charges. For example, the control signal may cause a threshold amount of current to flow through a heating element that causes detonation. When the one or more explosive charges detonate, the frangible bolt separates into two or more pieces. Positioning of the one or more explosive charges in the frangible bolt determines locations where the frangible bolt separates into pieces. Different types of separation fasteners may be used to couple the top plate **205** to the bottom plate **210** in various embodiments. For example, one or more frangible nuts are used to fasten the top plate **205** to the bottom plate **210**. The frangible nuts include explosive material configured to detonate in response to receiving a control signal, as further described above. In further embodiments, other types of separation fasteners that do not necessarily utilize explosives may be used to couple the top plate **205** to the bottom plate **210** that are capable of detaching the top plate **205** from the bottom plate **210** in response to receiving a control signal. For example, separable fasteners may be separable based on electronically actuated release mechanisms, materials that separate in response to temperature changes, or other separation mechanisms.

[0031] When the separation fasteners **315** are separated in a non-rollover condition, the top plate **205** remains coupled to the bottom plate **210** by the mounting cones **310** inserted into the pillow blocks **305** (based on force of gravity). However, the plates **205**, **210** will separate in a rollover condition when a portion of the top plate **205** has an appropriate orientation relative to a portion of the bottom plate **210**. Under rollover conditions, the top plate **205** is oriented relative to the bottom plate **210** to enable the one or more pillow blocks **305** to lift from corresponding mounting cones **310** on the bottom plate **210**. Hence, the mounting cones **310** and corresponding cavities of the pillow blocks **305** into which the mounting cones **310** are inserted operate as secondary securing mechanisms coupling the top plate **205** to the bottom plate **210**. This structure enables a semi-trailer to remain coupled to the semi-tractor even if the separation fasteners **315** are inadvertently separated (e.g., due to a fault condition in the sensor or controller) in the absence of an imminent rollover of the semi-trailer. However, during a rollover condition the top plate **205** pivots about a portion of the bottom plate **210** and causes one or more of the pillow blocks **305** to lift away from a corresponding mounting cone **310** of the bottom plate **210**, thereby causing the top plate **205** to separate from the bottom plate **210** of the rollover prevention device **200**. In various embodiments, the specific orientation of the portion of the top plate **205** to the portion of the bottom plate **210** that allows for separation may correspond to one or more rollover conditions maintained by the controller, as further described below (e.g., based on roll angle in combination with roll rate or other factors).

[0032] As further described below in conjunction with FIGS. **4** and **14**, a controller of the rollover prevention device **200** generates one or more control signals for causes separation of the separation fastener **315** (e.g., detonating frangible fasteners or initiating other separation mechanism). The controller determines whether to transmit a control signal to a separation fastener based on analysis of data captured by one or more sensors, as further described below in conjunction with FIGS. **4** and **14**. The controller may generate other data or capture other types of data in various embodiments.

[0033] FIG. **4** is a top perspective view of one embodiment of the rollover prevention device **200** without a top plate **205** installed. As shown in FIG. **4**, one or more pillow blocks **305** are coupled to the bottom plate **210** of the rollover prevention device **200**, as further described above in conjunction with FIG. **2**. The top plate **205** couples to the one or more pillow blocks **305**, as further described above in conjunction with FIG. **2**. Additionally, the rollover prevention device **200** shown in FIG. **4** has a controller **400** coupled to the bottom plate **210**. While FIG. **4** shows an embodiment with the controller **400** coupled to the bottom plate **210**, in other embodiments the controller **400** may be a discrete component positioned proximate to the bottom plate **210** and coupled to one or more of the separation fasteners **315**.

[0034] FIG. **5** illustrates a block diagram of one embodiment of a controller **400**. For purposes of illustration, the controller **400** is described as included in or coupled to a rollover prevention device **200**. However, in various embodiments, the controller **400** may instead be included in or coupled to a fifth wheel connector **105** that does not include one or more separable fasteners **315** for rollover prevention. Various functionality of the controller **400** unrelated to controlling a rollover prevention device **200** may therefore be provided in embodiments where the controller **400** is coupled to (or included in) a fifth wheel **105** without separable fasteners.

[0035] In various embodiments, the controller **400** includes an inertial measurement unit (IMU) **500** including one or more sensors. For example, the IMU **500** includes a three-axis accelerometer measuring acceleration in three orthogonal directions. In some embodiments, the IMU **500** includes a three-axis accelerometer and a three-axis gyroscope that determines angular velocity in one or more directions. Alternatively, the IMU **500** includes a gyroscope without an accelerometer. However, in other embodiments, the IMU **500** includes additional sensors, such as a magnetometer or a position sensor (e.g., a global positioning system (GPS) sensor). Multiple sensors may be included in the controller **400** in various embodiments. Further, in some embodiments, the IMU **500** is external to the controller **400** and communicatively coupled, through a wireless communication channel or a wired communication channel, to the controller **400**. Additional types of sensors may be included in the controller **400** or coupled to the controller **400** in various embodiments. In other embodiments, the IMU **500** is external to the controller **400** and is communicatively coupled to the controller **400**. Further in some embodiments, an additional IMU **550** is coupled to the controller **400**.

[0036] Based on acceleration measured in different directions by a multi-axis accelerometer (e.g., a multi-axis accelerometer included in the IMU **500**), a roll determination module **505** of the controller **400** determines a roll angle of the rollover prevention device **200**, or a roll angle of a position where a sensor including an accelerometer is positioned. In various embodiments, the controller **400** includes a processor that receives data from the IMU **500**, or from additional sensors, and determines the roll angle or the roll rate or the rollover prevention device **200**. In various embodiments, the determined roll angle is measured relative to a horizontal axis orthogonal to the sensor, providing a measure of a deviation from horizontal of the position of the sensor (e.g., at the rollover prevention device **200**). Additionally, the roll determination module **505** determines a roll rate of the rollover prevention device **200** based on data captured by the multi-axis accelerometer (e.g., included in the IMU **500**) or other sensor coupled to the controller **400**. The roll rate indicates a change in the roll angle relative to an axis of the rollover prevention device **200** (or of a sensor location) over time. For example, the roll rate indicates a rate at which the roll angle of the rollover prevention device **200** relative to a horizontal axis orthogonal to the sensor (e.g., the IMU **500**) over time. In various embodiments, the roll determination module determines the roll rate in degrees per second or in radians per second.

[0037] The controller **400** includes a rollover determination module **510** determines whether the roll angle or the roll rate satisfy a rollover condition. In various embodiments, the rollover determination module **510** stores a table of rollover conditions to which the roll angle and/or the roll rate are compared. In various embodiments, the rollover determination module **510** determines a rollover condition is satisfied when a determined roll angle is within a threshold amount or exceeds a threshold roll angle. The threshold roll angle is predetermined and stored by the rollover determination module **510** in some embodiments. In other embodiments, one or more rollover conditions comprise one or more combinations of threshold values of a combination of data measured by one or more sensors coupled to the controller or data determined by the controller based on data captured by one or more sensors. Different embodiments may include different types of data in one or more rollover conditions One or more rollover conditions may be stored in a non-volatile storage device, such as a solid state memory, in various embodiments. Alternatively, the rollover determination module **510** dynamically determines the threshold roll angle based on data captured by one or more sensors, as further described below. In response to determining the determined roll angle is within the threshold amount or exceeds the threshold roll angle, the rollover determination module **510** transmits a control signal to separation fastener **315** (such as frangible fasteners), of the rollover prevention device **200**. As further described above in conjunction with FIG. **2**, in response to receiving the control signal, in embodiments where the separation fasteners **315** comprise frangible bolts, explosive charges in the frangible bolts detonate to separate the frangible bolt into two or more pieces. Detonating the frangible bolt (or separating another separation fastener **315**) allows the top plate **205** of the rollover prevention device **200** coupled to a semi-tractor to detach from the bottom plate of the rollover prevention device **200** when the roll angle of the rollover prevention device **200** equals or exceeds the threshold roll angle.

[0038] In other situations, the rollover determination module **510** may determine that a rollover condition is satisfied in response to determining a roll rate of the rollover prevention device **200** equals or exceeds a threshold roll rate. For example, having a roll rate greater than a threshold rate, even when the roll angle is greater than the threshold amount from the threshold roll angle satisfies a rollover condition indicating an increased likelihood of a semi-trailer coupled to the top plate **205** of the rollover prevention device rolling over. To mitigate this risk, the table maintained by the rollover determination module **510** includes one or more rollover conditions comprising different roll angles, or ranges of roll angles, associated with threshold roll rates. In response to a roll rate determined by the roll determination module **505** equaling or exceeding a threshold rate associated with a roll angle determined by the roll determination module **505**, the rollover determination module **510** transmits a control signal to the separation fastener **315** to separate the separation fasteners **315** (e.g., to detonate an explosive charge in the frangible bolts), as further described above. Maintaining the table of threshold roll rates in association with roll angles allows determination of whether a rollover condition is satisfied based on different combinations of roll angle and roll rate that the rollover determination module **510** receives as inputs. This allows the separation fastener **315** to be separated (e.g., detonated) based on threshold roll rates that may vary for different ranges of roll angle, providing protection against a wider range of potential rollover conditions.

[0039] In some embodiments, the controller **400** includes a center of gravity module **515** that determines a center of gravity of a semi-trailer coupled to the rollover prevention device **200** and uses the center of gravity to as a factor in determining one or more rollover conditions for the semi-trailer based on how the semi-trailer is loaded. To determine the center of gravity of the semi-trailer, the controller **400** is coupled to one or more vertical load cells **520** included in the rollover prevention device **200**. A vertical load cell **520** determines a vertical load applied to the rollover prevention device **200** by a semi-trailer coupled to the rollover prevention device **200**. In various embodiments, a vertical load cell **520** is coupled to the controller **400** through a wired connection, while in other embodiments a vertical load cell **520** is wirelessly coupled to the controller **400**. Different types of vertical load cells **520** may be used in different embodiments. Hence, the controller **400** receives the vertical load applied to the rollover prevention device **200** by a semi-trailer coupled to the rollover prevention device **200** from the one or more vertical load cells **520**.

[0040] Alternatively, the controller **400** is coupled to one or more pressure sensors **525** mounted in a suspension of a semi-trailer coupled to the rollover prevention device **200** or is coupled to one or more pressure sensors **525** mounted in a suspension of a semi-tractor **100** coupled to the rollover prevention device **200**. The pressure sensors **525** may be positioned in the suspension of the semi-trailer proximate to the rollover prevention device **200** in various embodiments. A pressure sensor **525** determines a downward pressure applied by a load to the pressure sensors **525**, which is communicated to the controller **400** through a wireless communication channel or through a wired communication channel.

[0041] In other embodiments, the controller **400** captures oscillation of the rollover prevention device **200** through one or more accelerometers **530**, such as an accelerometer included in the IMU **500** or another accelerometer **530**. For example, the controller **400** captures vertical oscillation of the rollover prevention device **220** captured by one or more accelerometers **530**. However, in other embodiments, the controller **400** captures oscillations of the rollover prevention device **220** along one or more axes. For example, the controller **400** captures one or more of vertical oscillation, lateral oscillation, or longitudinal oscillation of the rollover prevention device **200** captured by one or more accelerometers. As another example, the controller **400** captures one or more of roll oscillation, pitch oscillation, or yaw oscillation captured by one or more gyroscopes or IMUs **500**, **550**. One or more of the accelerometers may be included in the controller **400** or one or more accelerometers **530** may be external to the controller **400** and communicatively coupled to the controller **400**. In some embodiments, one or more accelerometers **530** are included in the controller **400**, while one or more additional accelerometers **530** are external to the controller **400** and coupled to the controller **400**. Based on data from one or more sensors coupled to the controller (e.g., accelerometers **530**) the controller **400** determines a distribution of oscillation frequencies of the rollover prevention device **200** along one or more axes (e.g., a distribution of vertical oscillation frequencies). The controller **400** includes a vertical load module **535** that applies a trained machine-learning model to the distribution of oscillation frequencies, with the machine-learning model generating a vertical load applied to the rollover prevention device **200** from the distribution of oscillation frequencies. In various embodiments, the machine-learning model receives a Fourier transform of the distribution of oscillation frequencies as input and outputs a vertical load applied to the rollover prevention device **200** based on the Fourier transform. In various embodiments, the machine-learning model receives a distribution of oscillation frequencies of the rollover prevention device **200** along one or more axes, or one or more quantities based on one or more oscillation frequencies to determine a load applied to the rollover prevention device **200**, or another characteristic of the semi-tractor **100** or of the semi-trailer. For example, oscillation frequencies along six axes are received by the machine-learning model as input to determine one or more characteristics of the semi-tractor **100** or of the semi-trailer. In various embodiments, a the machine-learning model comprises a physics-based model fitted to a training dataset (e.g., a historical dataset of Fourier transforms of oscillation frequencies, a time history dataset of oscillation, etc.) obtained by a computing device coupled to the controller **400** to determine vertical load of another characteristic of the semi-tractor **100** or the semi-trailer dataset, whether, to determine vehicle characteristics such as vertical load.

[0042] In various embodiments, a computing device coupled to the controller **400**, trains the machine-learning model based on a training dataset including multiple training examples. Each training example includes a training distribution of oscillation frequencies (e.g., vertical oscillation frequencies, oscillation frequencies along multiple axes) and has a label indicating a vertical load corresponding to the training distribution of oscillation frequencies. To train the machine-learning model, a set of weights comprising the machine-learning model are initialized, and the machine-learning model is applied to each training example of the training dataset.

[0043] Applying the machine-learning model to multiple training examples updates the parameters (e.g., the weights) comprising the machine-learning model. The parameters comprising the machine-learning model transform the input data—the distribution of oscillation frequencies—into a vertical load applied to the rollover prevention device **200**. When applied to a training example, the machine-learning model generates a predicted vertical load applied to the rollover prevention device **200**.

[0044] For each training example to which the machine-learning model is applied, a computing device coupled to the controller **400** (e.g., a server coupled to the controller **400** via a network) generates a score comprising an error term based on the predicted vertical load applied to the rollover prevention device **200** and the label applied to the training example. The error term is larger when a difference between the predicted vertical load applied to the rollover prevention device **200** and the label applied to the training example is larger. Similarly, the error term is smaller when the difference between the predicted vertical load applied to the rollover prevention device **200** and the label applied to the training example is smaller. In various embodiments, the vertical load module **535**, or a computing device coupled to the controller **400**, generates the error term using a loss function based on a difference between the predicted vertical load applied to the rollover prevention device **200** and the label applied to the training example using a loss function. Example loss functions include a mean square error function, a mean absolute error, a hinge loss function, and a cross-entropy loss function.

[0045] A computing device coupled to the controller **400**, backpropagates the error term to update the set of parameters comprising the machine-learning model and stops backpropagation in response to the error term, or to the loss function, satisfying one or more criteria. For example, computing device backpropagates the error term through the machine-learning model to update parameters of the machine-learning model until the error term has less than a threshold value. For example, the computing device coupled to the controller **400**, may apply gradient descent to update the set of parameters. The computing device coupled to the controller **400**, stores the set of parameters comprising the machine-learning model on a non-transitory computer readable storage medium after stopping the backpropagation. Subsequently, the computing device transmits the trained machine-learning model to the controller **400**, which stores the trained machine-learning model in the vertical load module **535** for application. Alternatively, the machine-learning model may be stored in a cloud environment and may be accessed by the controller **400** via a network connection.

[0046] The vertical load module **535** may dynamically retrain the machine-learning model or transmit data to a computing device for retraining the machine-learning model over time. For example, the vertical load module **535** receives a measured vertical load for the rollover prevention device **200** that is associated with a distribution of sensed vertical oscillation frequencies. The vertical load module **535**, or the computing device coupled to the controller **400**, uses the combination of measured vertical load and distribution of sensed vertical oscillation frequencies as a training example for the machine-learning model to modify one or more parameters of the machine-learning model as further described above. In some embodiments, the machine-learning model is locally stored by the vertical load module **535**, which applies the machine-learning model to sensed vertical oscillation frequencies received by the one or more accelerometers **530**. Alternatively, the controller **400** transmits sensed vertical oscillation frequencies to a computing device via a wireless or a wired communication channel, the computing device applies the machine-learning model to the sensed vertical oscillation frequences, and transmits a vertical load applied to the rollover prevention device **200** to the controller **400** via the wireless or wired communication channel.

[0047] In various embodiments, the controller **400** also includes a mass determination module **540** configured to determines a mass of a semi-trailer coupled to the rollover prevention device **200**. In some embodiments, the controller **400** is coupled to one or more load cells **545** positioned in the rollover prevention device **200**. The load cells **545** measure a force used to accelerate a semi-trailer coupled to the rollover prevention device **200** during acceleration of the semi-trailer. Based on the measured force and a measured acceleration of the semi-trailer coupled to the rollover prevention device **200**, the mass determination module **540** determines the mass of the semi-trailer coupled to the rollover prevention device **200**. For example, the mass of the semi-trailer is determined as a ratio of the measured force to the measured acceleration of the semi-trailer. As further described above, the center of gravity module **515** determines a center of gravity of the semi-trailer based at least in part on the mass of the semi-trailer, and may account for the center of gravity of the semi-trailer in one or more rollover conditions in various embodiments.

[0048] In other embodiments, the mass determination module **540** additionally or alternatively determines a mass of a semi-trailer coupled to the rollover prevention device **200** based on vertical load applied to the rollover prevention device **200** for each of a plurality of positions of a suspension of the semi-trailer. For example, one or more vertical load cells **520** included in the rollover prevention device **200** and coupled to the controller **400** determine a first vertical load applied to the rollover prevention device **200** by a semi-trailer coupled to the rollover prevention device **200** when the suspension of the semi-trailer is in a first position. The one or more vertical load cells **520** included in the rollover prevention device **200** and coupled to the controller **400** also determine a second vertical load applied to the rollover prevention device **200** by a semi-trailer coupled to the rollover prevention device **200** when the suspension of the semi-trailer is in a second position. Based on the first vertical load, the second vertical load, and a distance between the first position of the suspension and the second position of the suspension, the mass determination module **540** determines the mass of the semi-trailer.

[0049] In various embodiments, the center of gravity module **515** may receive data from one or more sensors measuring a frequency of roll oscillations of the rollover prevention device **200** while a semi-tractor **100** coupled to the rollover prevention device **200** is in motion. In various embodiments, one or more accelerometers **530** included in the controller **400**, or coupled to the controller **400**, captures the frequency of roll oscillations. Alternatively or additionally, one or more gyroscopes or load cells **565** coupled to the controller **400**, or included in the controller **400** capture a frequency of roll oscillations of the rollover prevention device **200**. Based on the mass of the semi-trailer (which may be determined by the mass determination module **540**, as further described above) and the frequency of roll oscillations of the rollover prevention device **200**, the center of gravity module **515** determines a height of a center of gravity of the semi-trailer. In various embodiments, the rollover determination module **510** modifies one or more rollover conditions, such as a threshold roll angle or a threshold roll rate, based on the height of the center of gravity of the semi-trailer. For example, the rollover determination module decreases the threshold roll angle or the threshold roll rate in response to determining a height of the center of gravity of the semi-trailer is greater than a threshold height or in response to determining the height of the center of gravity of the semi-trailer increases by at least a threshold amount from a previously determined height. As another example, the rollover determination module increases the threshold roll angle and/or the threshold roll rate (or otherwise dynamically modifies the detection criteria for the rollover condition) in response to determining a height of the center of gravity of the semi-trailer is less than an additional threshold height or in response to determining the height of the center of gravity of the semi-trailer decreases by at least a threshold amount from a previously determined height.

[0050] Additionally, the controller **400** may transmit a warning signal to one or more devices in response to determining a height of the center of gravity of the semi-trailer equals or exceeds the threshold height. Determining the height of the center of gravity of the semi-trailer equals or exceeds the threshold height equals or exceeds the threshold height indicates loading of cargo in the semi-trailer increases a likelihood of the semi-trailer rolling over while in motion. The warning signal may be transmitted from the controller **400** to a client device in the semi-tractor **100** (e.g., a client device of a driver of the semi-tractor **100**, a computing device included in the semi-tractor **100**, etc.). Additionally or alternatively, the warning signal may be transmitted from the controller **400** to a remote client device, such as a client device of an entity associated with the semi-tractor **100**. The warning signal provides a driver or other entity with advanced warning of an increased risk of the semi-trailer rolling over, allowing a driver to take preventive action to mitigate the risk of the semi-trailer rolling over (e.g., reconfiguring cargo in the semi-trailer, altering one or more driving characteristics, etc.).

[0051] In various embodiments, based on the vertical load applied to the rollover prevention device **200** by a semi-trailer coupled to the rollover prevention device **200**, the mass of the semi-trailer coupled to the rollover prevention device **200**, and a specified weight of the semi-trailer without cargo, the mass determination module **540** determines a mass of the semi-trailer on each axle of the semi-tractor **100** or each axle of the semi-trailer. Determining the mass of the semi-trailer on each axle of the semi-tractor **100** or of the semi-trailer allows the mass determination module **540** to determine whether one or more portions of the suspension supporting the semi-trailer are capable of being repositioned. For example, the mass determination module maintains a threshold load on various axles of the suspension supporting the semi-trailer, and determines whether a load on an axle of the semi-trailer (or an axle of the semi-tractor **100**) is less than the threshold mass when the axle when the suspension of the semi-trailer is in a specific position. The controller **400** prevents repositioning the suspension to the specific position in response to the load on the axle of the semi-trailer (or the axle of the semi-tractor **100**) equaling or exceeding the threshold mass. As further described below in conjunction with FIG. **11**, the rollover prevention device **200** may be repositioned along a portion of the semi-tractor **100** based on one or more movement signals from the controller **400** in various embodiments. Repositioning the rollover prevention device **200** repositions a portion of the suspension of the semi-trailer, and the controller **400** prevents repositioning of the semi-trailer suspension to a position where a load on an axle of the semi-trailer suspension (or a mass of an axle of the semi-tractor **100** suspension) equals or exceeds the threshold mass.

[0052] While FIG. **4** shows an example rollover prevention device **200** including the controller **400**, in alternative embodiments, the controller **400** is included in or coupled to a fifth wheel **105** configured to couple a semi-trailer to a semi-tractor **100**, such as a fifth wheel **105** that does not include one or more frangible fasteners. When included in or coupled to a fifth wheel **105**, the controller **400** includes one or more of the IMU **500**, the roll determination module **505**, the center of gravity module **515**, the vertical load module **535**, and the mass determination module **540**. Similarly, a controller included in a fifth wheel **105** may be coupled to one or more vertical load cells **520**, one or more pressure sensors **525**, one or more accelerometers **530**, and one or more load cells **545**.

[0053] A controller **400** included in a fifth wheel **105** or coupled to a fifth wheel **105** determines a center of gravity of a semi-trailer coupled to the fifth wheel **105** and may generate a warning signal or other message in response to the center of gravity satisfying one or more conditions. The warning may be transmitted to a client device of a driver of the semi-tractor or to a computing device of an entity associated with the semi-tractor. As further described above, the center of gravity module **515** of a controller included in or coupled to the fifth wheel **105** determines the center of gravity of the semi-trailer coupled to the fifth wheel **105** based on data from one or more vertical load cells **520** included in the fifth wheel **105** that determine vertical load applied to the fifth wheel **105**.

[0054] Alternatively, the controller **400** is coupled to one or more pressure sensors **525** mounted in a suspension of a semi-trailer coupled to the fifth wheel **105** or is coupled to one or more pressure sensors **525** mounted in a suspension of a semi-tractor **100** coupled to the fifth wheel **105**. The pressure sensors **525** may be positioned in the suspension of the semi-trailer proximate to the fifth wheel **105** in various embodiments. A pressure sensor **525** determines a downward pressure applied by a load to the pressure sensors **525**, which is communicated to the controller **400** through a wireless communication channel or through a wired communication channel.

[0055] In other embodiments, the controller **400** captures oscillation of the fifth wheel **105** through one or more sensors, as further described above. Based on data from the one or more accelerometers **530**, the controller **400** determines a distribution of oscillation frequencies of the fifth wheel **105** and determines one or more characteristics of the semi-tractor **100** or the semi-trailer through application of a trained machine-learning model to the distribution of oscillation frequencies, as further described above.

[0056] As further described above, the controller **400** also includes a mass determination module **540** configured to determine a mass of a semi-trailer coupled to the fifth wheel **105**. In some embodiments, the controller **400** is coupled to one or more load cells **545** positioned in the fifth wheel **105**. The load cells **545** measure a force used to accelerate a semi-trailer coupled to the fifth wheel **105** during acceleration of the semi-trailer. Based on the measured force and a measured acceleration of the semi-trailer coupled to the fifth wheel **105**, the mass determination module **540** determines the mass of the semi-trailer coupled to the fifth wheel **105**. For example, the mass of the semi-trailer is determined as a ratio of the measured force to the measured acceleration of the semi-trailer. As further described above, the center of gravity module **515** determines a center of gravity of the semi-trailer based at least in part on the mass of the semi-trailer, and may account for the center of gravity of the semi-trailer in one or more rollover conditions in various embodiments.

[0057] In other embodiments, the mass determination module **540** additionally or alternatively determines a mass of a semi-trailer coupled to the fifth wheel **105** based on vertical load applied to the fifth wheel **105** for each of a plurality of positions of a suspension of the semi-trailer. For example, one or more vertical load cells **520** included in the fifth wheel **105** and coupled to the controller **400** determine a first vertical load applied to the fifth wheel **105** by a semi-trailer coupled to the fifth wheel **105** when the suspension of the semi-trailer is in a first position. The one or more vertical load cells **520** included in the fifth wheel **105** and coupled to the controller **400** also determine a second vertical load applied to the fifth wheel **105** by a semi-trailer coupled to the fifth wheel **105** when the suspension of the semi-trailer is in a second position. Based on the first vertical load, the second vertical load, and a distance between the first position of the suspension and the second position of the suspension, the mass determination module **540** determines the mass of the semi-trailer.

[0058] In various embodiments, the center of gravity module **515** may receive data from one or more sensors measuring a frequency of roll oscillations of the fifth wheel **105** while a semi-tractor **100** coupled to the fifth wheel **105** is in motion. In various embodiments, one or more accelerometers **530** included in the controller **400**, or coupled to the controller **400**, captures the frequency of roll oscillations. Alternatively or additionally, one or more gyroscopes or load cells **545** coupled to the controller **400**, or included in the controller **400** capture a frequency of roll oscillations of the rollover prevention device **200**. Based on the mass of the semi-trailer (which may be determined by the mass determination module **540**, as further described above) and the frequency of roll oscillations of the fifth wheel **105**, the center of gravity module **515** determines a height of a center of gravity of the semi-trailer. In various embodiments, the controller **400** transmits a notification to a client device of a driver of the semi-tractor **100** in response to determining the height of the center of gravity of the semi-trailer equals or exceeds a threshold height. Alternatively, or additionally, the controller **400** transmits a notification to a client device of an entity associated with the semi-tractor **100** in response to determining the height of the center of gravity of the semi-trailer equals or exceeds a threshold height.

[0059] Hence, the controller **400** may be used in combination with a fifth wheel **105** and to determine a height of a center of gravity of a semi-trailer coupled to the fifth wheel **105** as, further described above, providing information about loading of the semi-trailer to a driver of a semi-tractor **100** or to another entity. Similarly, when used in combination with a fifth wheel **105**, the controller **440** may determine a mass of a semi-trailer coupled to the fifth wheel **105** or a mass applied to one or more axles of the semi-trailer or the semi-tractor **100**, which may be used to provide notifications to a driver of the semi-tractor **100** or to reposition the semi-trailer relative to the semi-tractor, as further described below in conjunction with FIGS. **11-13**B. Hence, various functionalities further described above may be provided by the controller **400** when coupled to, or included in, a fifth wheel **105** that does not include one or more frangible fasteners.

[0060] FIG. **6** is a side cross-section view of an example of a rollover prevention device **200** without a top plate **205** installed. As shown in FIG. **6**, one or more mounting cones **310** of the bottom plate **210** of the rollover prevention device **200** are inserted into cavities of a pillow block **305**. In the example of FIG. **6**, the pillow block **305** includes two cavities, with a mounting cone **310** from the bottom plate **210** inserted into each cavity. As further described above in conjunction with FIG. **3**, the mounting cones **310** are rigidly attached to the bottom plate **210**.

[0061] Additionally, the pillow block **305** is coupled to the bottom plate **210** through one or more separation fastener **315**. As further described above in conjunction with FIG. **3**, a separation fastener **315** is inserted through the mounting cone **310** into a portion of a pillow block **305** having a cavity into which the mounting cone **310** is inserted **310**. The cavity of the pillow block **305** includes threads configured to interlock with threads of the separation fastener **315** to couple the pillow block **305** to the bottom plate **210**. Hence, the pillow block **305** couples the top plate **205** to the bottom plate **210**.

[0062] FIG. **7** illustrates an example of a top plate **205** of the rollover prevention device **200** pivoting off of the mounting cones **310** to detach from the bottom plate **210** of the rollover prevention device **200**. For example, the top plate **205** pivots about a pivot point **700** on the bottom plate **120**. The pivot point **700** is a mounting cone **310** of the bottom plate **210** in various embodiments. As shown in FIG. **7**, a side of the top plate **205** distal from the pivot point **700** raises vertically from the bottom plate **210** so a pillow block **305** nearest to the side of the top plate **205** raising vertically from the bottom plate **210** is above the mounting cone **310** nearest to the side of the top plate **205** raising vertically. This detaches the side of the top plate **205** from the bottom plate **210**. The top plate **205** may continue to pivot about the pivot point **700** so a pillow block **305** nearest to the pivot point **700** raises above the mounting cone **310** nearest to the pivot point **800** to detach the top plate **205** from the bottom plate **210**.

[0063] FIG. **8** illustrates one embodiment of assembling pillow blocks **305** of the rollover prevention device **200** onto mounting cones **310** of the rollover prevention device **200**. As shown in FIG. **8**, a mounting cone **310** is inserted into a cavity of the pillow block **305**. In various embodiments, a portion of the cavity of the pillow block **305** includes threads, and a separation fastener **315** is threaded through the mounting cone **310** into the cavity of the pillow block **305** so threads of the separation fastener **315** interlock with the threads of the cavity of the pillow block **305**.

[0064] FIG. **9** is a top perspective view of an alternative embodiment of a pillow block **305** and a bottom plate **210** of a rollover prevention device **200**. In the embodiment shown by FIG. **9**, each pillow block comprises a pillow block top component **900** coupled to a pillow block bottom component **905**. For example, a fastener couples to the pillow block top component **900** and to the pillow block bottom competent **905** to couple the pillow block top component **900** to the pillow block bottom component **905**. Additionally, the pillow block top component **900** and the pillow block bottom component **905** interlock with each other to prevent forward motion of a semi-trailer coupled to a top plate **205** of the rollover prevention device **200** towards a semi-tractor **100** coupled to the bottom plate **210** of the rollover prevention device **200**. For example, a hinge couples the pillow block top component **900** to the pillow block bottom component **805**, a further described below in conjunction with FIG. **10**. The pillow block top component **900** is configured to be coupled to the top plate **205**, while the pillow block bottom component **905** is configured to be coupled to the bottom plate **210**.

[0065] FIG. **10** is a side cross-sectional view of the alternative embodiment of the pillow block **305** and the bottom plate **210** of a rollover prevention device **200** shown by FIG. **9**. As shown by FIG. **10**, the pillow block top component **900** is coupled to the pillow block bottom component **905** by a hinge **1000**. The hinge **1000** prevents the pillow block top component **900** from separating from the pillow block bottom component **905** unless the pillow block top component **900** is lifted from the pillow block bottom component **905** or unless the pillow block top component **900** is pivoted upward from the pillow block bottom component **905**.

[0066] Additionally, a separation fastener **315** is inserted horizontally through the pillow block through a portion of the pillow block top component **900** and a portion of the pillow block bottom component **905**. The separation fastener **315** extends through the pillow block to at least partially contact the hinge **1000**, so the separation fastener **315** secures the hinge **1000**. As further described above in conjunction with FIG. **3**, in various embodiments a separation fastener **315** comprises a frangible bolt including one or more explosive charges that detonate in response to receiving a control signal. When the separation fastener **315** separates (e.g., a frangible bolt detonates), the pillow block top component **900** separates from the pillow block bottom component **905** when rollover conditions are met (e.g., when a roll angle of the rollover prevention device **200** equals or exceeds a threshold roll angle and/or when a roll rate of the rollover prevention device **200** equals or exceeds a threshold roll rate). When the rollover condition is met, the top plate **205** can pivot about a portion of the bottom plate **210** so the hinge **1000** of a pillow block separates. This detaches the pillow block top component **900** from the pillow block bottom component **905**, enabling the top plate **205** to detach from the bottom plate **210** of the rollover prevention device **200**.

[0067] FIG. **11** is an overhead view of one embodiment of a repositionable fifth wheel connector **1150**. In the example of FIG. **11**, the repositionable fifth wheel connector **1150** includes, or is coupled to, an actuator **1100** that enables repositioning of the fifth wheel connector **1150** relative to a guiding system **1105** configured to be coupled to a surface of a semi-tractor **100**. The fifth wheel connector **1150** may comprise a rollover prevention device **200** that includes separable fasteners **315** as described above or may include a fixed fifth wheel connector that does not necessarily include separable fasteners **315** for rollover prevention.

[0068] The actuator **1100** receives one or more movement signals from a controller **400** to control positioning. In response to a movement signal, the actuator **1100** repositions the fifth wheel connector **1150** along the guiding system **1105**. For example, a first movement signal moves the fifth wheel connector **1150** along the guiding system **1105** in a direction towards the semi-tractor **100**. Similarly, a second movement signal moves the fifth wheel connector **1150** along the guiding system **1105** in a direction away from the semi-tractor **100**. In various embodiments, the actuator **1100** comprises a lead screw or a ball screw with an electric motor or a hydraulic motor that receives one or more movement signals from the controller **400**. Alternatively, the actuator **1100** comprises a rack and pinion system with an electric motor or a hydraulic motor that receives one or more movement signals from the controller **400**. In other embodiments, the actuator **1100** comprises a timing belt, a cable system, or a winch system having an electric motor or a hydraulic motor that receives one or more movement signals from the controller **400**.

[0069] In various embodiments, repositioning the fifth wheel connector **1150** along the guiding system **1105** via the actuator **1100** may be used in combination with control of wheel brakes to reposition the suspension of the semi-trailer relative to the semi-tractor **100**. By repositioning the suspension of the semi-trailer, mass is redistributed between axles of the semi-trailer suspension and axles of the semi-tractor suspension to comply with axle load specifications or other factors. For example, moving the fifth wheel connector **1150** in a first direction along the guiding system **1105** decreases load on an axle of the suspension of the semi-trailer and increases load on an axle of the suspension of the semi-tractor **100**. As another example, moving the fifth wheel connector **1150** in a second direction along the guiding system **1105** increases load on an axle of the suspension of the semi-trailer and decreases load on an axle of the suspension of the semi-tractor **100**.

[0070] In various embodiments, when repositioning the actuator **1100**, the controller **400** determines a current and an instantaneous voltage applied to the actuator **1100** to reposition the fifth wheel connector **1150** along the guiding system **1105**. From the current and a force conversion factor for the actuator **1100**, the controller **400** determines a force applied by the actuator **1100** and determines the mass of the semi-trailer from the determined force. As further described above in conjunction with FIG. **5**, the controller **400** determines a center of gravity of the semi-trailer based at least in part on the mass of the semi-trailer. In embodiments equipped with rollover prevention, the center of gravity of the semi-trailer may be utilized, in part, to configure rollover detection conditions and determine when to release the separation fastener **315**. Hence, the rollover conditions may change as the fifth wheel connector **1150** is moved to different positions along the guiding system **1105**.

[0071] Further, repositioning the suspension of the semi-trailer may be based on regulations for regions where the semi-tractor **100** is moving. For example, compliance with different state regulation may involve adjusting locations of semi-trailer suspension relative to the semi-tractor **100** or relative to an end of the semi-trailer. The controller **400** may operate to enable repositioning the fifth wheel connector **1150** to a particular position satisfying a regulation for a region where the semi-tractor **100** is moving. In various embodiments, the controller **400** is coupled to a sensor that determines a position of the semi-trailer suspension. For example, the sensor determining the position of the semi-trailer suspension is a camera, a time-of-flight sensor (e.g., a laser time-of-flight sensor), an ultrasonic sensor, or another sensor capable of determining a position of the suspension of the semi-trailer.

[0072] In some embodiments, the controller **400** transmits a movement signal to the actuator **1100** based on a speed of the semi-tractor **100** to which the fifth wheel connector **1150** is coupled. For example, the controller **400** determines or receives a captured speed of the semi-tractor **100** to which the fifth wheel connector **1150** is coupled, and generates a movement signal for the actuator **1100** based at least in part on the speed. For example, the controller **400** generates a movement signal causing the actuator **1100** to move the fifth wheel connector **1150** nearer to the semi-tractor **100** in response to determining the speed equals or exceeds a threshold or in response to determining the speed has increased relative to a prior speed, to reduce drag from airflow over the semi-tractor **100** and the semi-trailer during movement. As another example, the controller **400** generates an alternative a movement signal causing the actuator **1100** to move the fifth wheel connector **1150** farther from the semi-tractor **100** in response to determining the speed is less than an additional threshold or in response to determining the speed has decreased relative to a prior speed, to increase maneuverability of the semi-tractor **100**.

[0073] In various embodiments, one or more movement signals generated by the controller **400** are also based on a load on an axle of the semi-trailer and a load on an axle of the semi-tractor **100** when the suspension of the semi-trailer is in a particular position. For example, in response to determining an updated position of the suspension of the semi-trailer based on a speed of semi-tractor coupled to the semi-trailer, the controller **400** determines whether the updated position of the suspension of the semi-trailer results in a load on the axle of the semi-tractor **100** that equals or exceeds a threshold value. Determination of the load on one or more axles is further described above in conjunction with FIG. **5**. In response to the updated position of the suspension of the semi-trailer resulting in a load on the axle of the semi-tractor **100** that equals or exceeds a threshold value, the controller **400** determines an alternative position of the suspension of the semi-trailer resulting the load on the axle of the semi-tractor **100** being less than the threshold value and transmits a movement signal to the actuator **1100** based on the updated position. This prevents the controller **400** from repositioning the suspension of the semi-trailer to a position where the load on the axle of the semi-tractor **100** exceeds a threshold.

[0074] FIG. **12** is an overhead view of an alternative embodiment of a repositionable fifth wheel connector **1150**. The fifth wheel connector **1150** is coupled to a guiding system **1105** as described above to enable repositioning. For example, a bottom plate of the fifth wheel connector **1150** is coupled to the guiding system **1105**. The guiding system **1105** is configured to be coupled to a surface of a semi-tractor **100**.

[0075] The guiding system **1105** includes one or more lead screws **1200** coupled to the fifth wheel connector **1150**. For example, the bottom plate of the fifth wheel connector **1150** includes one or more threaded holes, and a lead screw **1200** is inserted into a corresponding threaded hole. Each lead screw **1200** is coupled to an actuator **1205**. Each lead screw **1200** may be coupled to a corresponding discrete actuator **1205** in some embodiments, while in other embodiments a common actuator is coupled to a common actuator **1205**. In various embodiments, the actuator **1205** comprises an electric motor or a hydraulic motor that receives one or more movement signals from the controller **400**. An actuator **1205** receives one or more movement signals from the controller **400** of the fifth wheel connector **1150**. In the example of FIG. **12**, each lead screw **1200** is coupled to an end of the guiding system **1105** opposite an end of the guiding system **1105** proximate to the actuator **1205**, so a lead screw **1200** spans a length of the guiding system **1105** and runs underneath the fifth wheel connector **1150**.

[0076] In response to a movement signal, the actuator **1205** repositions the fifth wheel connector **1150** along the guiding system **1105**. For example, a first movement signal causes the actuator **1205** to rotate a lead screw **1200** clockwise, which moves the fifth wheel connector **1150** along the guiding system **1105** in a first direction relative to the semi-tractor **100**. Similarly, a second movement signal causes the actuator **1205** to rotate a lead screw **1200** counterclockwise, which moves the fifth wheel connector **1150** along the guiding system **1105** in a second direction relative to the semi-tractor **100**, with the second direction opposite to the first direction. As described above, the fifth wheel connector **1105** in this embodiment may include a rollover prevention **200** with separable fasteners **315** or a fixed fifth wheel connector that does not necessarily include separable fasteners **315**.

[0077] FIGS. **13**A and **13**B are perspective views of another embodiment of a repositionable fifth wheel connector **1150**. In the example of FIGS. **13**A and **13**B, the fifth wheel connector **1150** is coupled to a guiding system **1105**. For example, a bottom plate of the fifth wheel connector **1150** is coupled to the guiding system **1105**. The guiding system **1105** is configured to be coupled to a surface of a semi-tractor **100**.

[0078] The guiding system **1105** includes one or more actuators **1300** coupled to one or more lead screws **1305**. In various embodiments, the lead screws **1305** are inserted through threaded holes in a stop **1310** positioned at a first end **1315** of the guiding system **1105**. The actuator **1300** comprises an electric motor or a hydraulic motor that receives one or more movement signals from the controller **400** in various embodiments. Each lead screw **1305** may be coupled to a corresponding discrete actuator **1300** in some embodiments, while in other embodiments a common actuator **1300** is coupled to each lead screw **1305**. In the embodiment shown by FIG. **13**A, a lead screw **1305** is coupled to a portion of the fifth wheel connector **1150** at a first end. For example, a first end of a lead screw **1305** is coupled to a portion of a bottom plate **210** of the fifth wheel connector **1150**.

[0079] In response to a movement signal, the actuator **1300** rotates the lead screw **1305** to move the fifth wheel connector **1150** along the guiding system **1105** relative to the stop **1310**. In the example of FIG. **13**A, in response to a first movement signal, the actuator **1300** actuator **1300** rotates a lead screw **1305** in a first direction (e.g., counterclockwise) to move the fifth wheel connector **1150** in a first direction relative to the stop **1310**, such as towards the stop **1310** in the example of FIG. **13**A. Similarly, FIG. **13**B shows the actuator **1300** rotating a lead screw **1305** in a second direction (e.g., clockwise) in response to a second movement signal. In response to the lead screw **1305** rotating in the second direction, the fifth wheel connector **1150** moves in a second direction relative to the stop **1310**, such as away from the stop **1310** in the example of FIG. **13**B. In the example of FIGS. **13**A and **13**B, the lead screws **1305** do not move underneath the fifth wheel connector **1150**, reducing an amount of space occupied by the guiding system **1105**. Further, FIGS. **13**A and **13**B show the stop **1310** and the first end **1315** of the guiding system **1105** opposite to the a horseshoe-shaped opening of the fifth wheel connector **1150** through which a kingpin protruding from a bottom front surface of a semi-trailer, preventing the actuator **1300** or a lead screw **1305** from being contacted when a semi-trailer is coupled to the fifth wheel connector **1150**.

[0080] In FIGS. **11-13**B the illustrated fifth wheel connector **1150** may include a rollover prevention device **200** that can be repositioned along the guiding system **1105**, or may comprise a fifth wheel **105** without separable fasteners **315**. These embodiments allow a fifth wheel connector **1150** with or without separable fasteners to be repositioned relative to a semi-tractor **100** in response to control signals from a controller **400**, as further described above. For example, a fifth wheel connector **1150** coupled to the guiding system **1105** is repositioned along the guiding system in response to one or more movement signals from a controller **400** determined based on a load on an axle of a semi-tractor **100** coupled to the fifth wheel connector **1150**, as further described above. As another example, a fifth wheel connector **1150** coupled to the guiding system **1105** is repositioned along the guiding system in response to one or more movement signals from a controller **400** determined based on a height of a center of gravity of a semi-trailer coupled to the fifth wheel **105**, as further described above.

[0081] FIG. **14** is a flowchart of an example embodiment of a method for determining whether to transmit a control signal to frangible fasteners of a rollover prevention device **200**. In various embodiments, steps of the method are performed by the controller **400** of the rollover prevention device **200**. Further, in various embodiments, the method may include different or additional steps than those described in conjunction with FIG. **14**. Additionally, in some embodiments, the steps of the method may be performed in a different order than the order described in conjunction with FIG. **14**.

[0082] The controller **400** determines **1405** a roll angle of the rollover prevention device **200**, as further described above in conjunction with FIG. **5**. In various embodiments, the controller **400** includes an inertial measurement unit including one or more sensors that capture a roll angle of the rollover prevention device **200** and the controller **400** receives the roll angle. As another example, the rollover prevention device **200** determines **1405** the roll angle based on data captured by one or more of the sensors. Example sensors include a three-axis accelerometer or a gyroscope, although other sensors or combinations of sensors may be used. As further described above in conjunction with FIG. **5**, in some embodiments, the one or more sensors are included in the controller, while in other embodiments the sensors are coupled to the controller **400**. The roll angle is measured relative to a horizontal axis orthogonal to the sensor, providing a measure of a deviation of the position of the sensor (e.g., at the rollover prevention device **200**) from the horizontal axis.

[0083] Additionally, the controller **400** determines **1410** a roll rate of the rollover prevention device **200**. One or more of the sensors capturing data used to determine **1205** the roll angle also capture data for determining **1410** the roll rate or the rollover prevention device **200**. The roll rate indicates a rate at which the roll angle of the rollover prevention device **200** relative to a horizontal axis orthogonal to the sensor (e.g., the IMU) changes over time. In various embodiments, the controller **400** determines **100** the roll rate in degrees per second or in radians per second.

[0084] The controller **400** determines **1415** whether the roll angle and/or the roll rate satisfy a rollover condition. In an embodiment, the controller **400** maintains a table that maps different combinations of roll angles and roll rates to an output indicating whether a rollover condition is met. The rollover combination may be based on the roll angle alone exceed a threshold, the roll rate alone exceeding a threshold, or a combination of roll angle and roll rate meeting a rollover condition. As further described above in conjunction with FIG. **5**, in various embodiments, one or more rollover conditions may be further based on a height of a center of gravity that the controller **400** determines for a semi-trailer coupled to the rollover prevention device **200**, so the controller **400** may dynamically adjust one or more rollover conditions based on characteristics of the semi-trailer.

[0085] In response to determining **1415** the roll angle and/or the roll rate satisfies a rollover condition, the controller **400** transmits **1220** a control signal to one or more separation fasteners **315** (e.g., frangible bolts) included in the rollover prevention device **200**. The frangible fasteners couple the top plate **205** of the rollover prevention device **200** to the bottom plate **210** of the rollover prevention device **200**. In response to receiving the control signal, one or more explosive charges in a frangible bolt detonate, separating the frangible fastener into multiple pieces (or a separation fastener **315** otherwise separates into multiple pieces). Separation of the frangible fastener allows a top plate **205** of the rollover prevention device **200** to detach from the bottom plate **210** of the rollover prevention device **200**, as further described above in conjunction with FIGS. **3** and **7**.

[0086] As further described above in conjunction with FIGS. **3** and **10**, in various embodiments the top plate **205** of the rollover prevention device **200** is coupled to the bottom plate **210** of the rollover prevention device **200** through one or more secondary securing mechanisms. Examples of secondary securing mechanisms include the mounting cones **310** and corresponding cavities of the pillow block in FIGS. **2** and **3** or the hinge **1000** coupling a pillow block top component **900** to a pillow block bottom component **905**, as further described above in conjunction with FIGS. **9** and **10**. When the frangible fasteners are detonated in the absence of a rollover condition (e.g., detonated due to a faulty sensor or controller detection), the secondary securing mechanisms may operate to maintain coupling the top plate **205** of the rollover prevention device **200** to the bottom plate **210** of the rollover prevention device **200** unless the top plate **205** rotates relative to the bottom plate **210** as during a rollover. For example, the secondary securing mechanisms couple the top plate **205** to the bottom plate **210** until one or more portions of the top plate **205** are raised above corresponding portions of the bottom plate **210**, as further described above in conjunction with FIG. **7**.

[0087] Under normal operation, in response to determining **1415** the determined roll angle and the determined roll rate do not satisfy at least one rollover condition, the controller **400** does not cause separation of the separable fastener **315**. The controller **400** continues to monitor the roll angle and roll rate, as further described above continuously or on a periodical interval.

[0088] The foregoing description of the embodiments has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the embodiments to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

[0089] The language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope is not limited by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments is intended to be illustrative, but not limiting, of the scope of the invention.

Claims

1. A rollover prevention device comprising: a top plate; a bottom plate coupled to the top plate by one or more separation fasteners; and a controller coupled to one or more sensors, the controller configured to: determine a roll angle of the rollover prevention device based on data from the one or more sensors; determine a roll rate of the rollover prevention device based on data from the one or more sensors; and transmit a control signal to each of the separable fasteners in response to determining at least one of the roll angle or the roll rate satisfies at least one rollover condition, the control signal separating one or more separation fasteners to enable the top plate to detach from the bottom plate.
2. The rollover prevention device of claim 1, wherein the one or more separation fasteners comprise frangible fasteners including one or more explosive charges, and wherein the one or more explosive charges in a frangible fastener detonate in response to the control signal to enable the top plate to detach from the bottom plate.
3. The rollover prevention device of claim 1, further comprising: one or more secondary securing mechanisms coupling the bottom plate to the top plate, the one or more secondary securing mechanisms coupling the top plate to the bottom plate after the one or more separation fasteners separate until a portion of the top plate has a specific orientation relative to a portion of the bottom plate.
4. The rollover prevention device of claim 3, wherein a secondary securing mechanism comprises a pillow block to which the top plate is coupled, the pillow block including a cavity into which a mounting cone coupled to the bottom plate is inserted.
5. The rollover prevention device of claim 4, wherein a separation fastener is inserted through the mounting cone into the cavity of the pillow block.
6. The rollover prevention device of claim 3, wherein a secondary securing mechanism comprises a pillow block having a pillow block top component coupled to the top plate and a pillow block bottom component coupled to the bottom plate, the pillow block top component coupled to the pillow block bottom component by a hinge.
7. The rollover prevention device of claim 6, wherein a separation fastener is inserted horizontally through the pillow block through a portion of the pillow block top component and the pillow block bottom component.
8. The rollover prevention device of claim 1, wherein determining the roll angle or the roll rate satisfies at least one rollover condition comprises: determining the roll angle is within a threshold amount of a threshold roll angle.
9. The rollover prevention device of claim 1, wherein determining the roll angle or the roll rate satisfies at least one rollover condition comprises: determining the roll angle is greater than a threshold amount of a threshold roll angle and the roll rate is greater than a threshold roll rate.
10. The rollover prevention device of claim 9, wherein the threshold roll rate is based on the roll angle.
11. The rollover prevention device of claim 1, wherein the top plate is configured to be coupled to a semi-trailer and the bottom plate is configured to be coupled to a semi-tractor.
12. The rollover prevention device of claim 11, wherein the controller is further configured to determine a height of a center of gravity of the semi-trailer based on data captured by one or more sensors coupled to the controller.
13. The rollover prevention device of claim 12, wherein one or more of the rollover conditions are based at least in part on the height of the center of gravity of the semi-trailer.
14. The rollover prevention device of claim 13, wherein the controller decreases a threshold roll angle in response to determining the height of the center of gravity of the semi-trailer is greater than a threshold value.

15. The rollover prevention device of claim 1, further comprising: a guiding system couple to the bottom plate; and an actuator coupled to the bottom plate and to the guiding system, the actuator configured to receive one or more movement signals from the controller and to move the bottom plate along the guiding system.

16. A rollover prevention device comprising: a top plate; a bottom plate coupled to the top plate by one or more separation fasteners; a guiding system couple to the bottom plate; an actuator coupled to the bottom plate and to the guiding system, the actuator configured move the bottom plate along the guiding system in response to receiving a movement signal; and a controller coupled to one or more sensors, the controller configured to: determine a roll angle of the rollover prevention device based on data from the one or more sensors; determine a roll rate of the rollover prevention device based on data from the one or more sensors; transmit a control signal to each of the one or more separation fasteners in response to determining at least one of the roll angle or the roll rate satisfies at least one rollover condition, the control signal separating one or more separation fasteners to enable the top plate to detach from the bottom plate; and generate a movement signal for the actuator based on data captured by the one or more sensors.

17. The rollover prevention device of claim 16, wherein the one or more separation fasteners comprise frangible fasteners including one or more explosive charges, and wherein the one or more explosive charges in a frangible fastener detonate in response to the control signal to enable the top plate to detach from the bottom plate.

18. The rollover prevention device of claim 16, wherein the top plate is configured to be coupled to a semi-trailer and the guiding system is configured to be coupled to a semi-tractor.

19. The rollover prevention device of claim 18, wherein the controller is further configured to: determine a speed of the semi-tractor; determine a load on an axle of the semi-trailer and a load on an axle of the semi-tractor when a suspension of the semi-trailer is in a position determined by the controller; determine an updated position of the suspension of the semi-trailer based on the speed of the semi-tractor; determine whether a load on the axle of the semi-tractor when the suspension of the semi-trailer is in an updated position equals or exceeds a threshold value; and generate a movement signal for the actuator to move the bottom plate along the guiding system so the suspension of the semi-trailer is in the updated position in response to determining the load on the axle of the semi-tractor when the suspension of the semi-trailer is in an updated position is less than the threshold value.

20. The rollover prevention device of claim 19, wherein the controller is further configured to: determine an alternative position of the suspension of the semi-trailer based on the speed of the semi-tractor in response to determining the load on the axle of the semi-tractor when the suspension of the semi-trailer is in an updated position equals or exceeds a threshold value.

21-25. (canceled)
